Lab report for Franck–Hertz experiment

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Aim: To show the existence of quantized energy levels in atoms by performing Franck–Hertz experiment

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1 Abstract

The Franck-Hertz experiment played a pivotal role in the development of quantum mechanics. This experiment provided crucial insights into the discrete nature of energy levels in atoms and demonstrated the quantization of electron energy states. This lab report delves into the experimental setup, theoretical underpinnings, and methodology of the Franck-Hertz experiment and concludes by analyzing and interpreting the results.

2 Apparatus used

1. Argon filled tetrode
2. Continuously variable filament power supply : 2.6 - 3.3 V
3. Continuously variable power supply for VG1K : 1.3 – 5 V
4. Continuously variable power supply for VG2A : 1.3 – 15 V
5. Continuously variable power supply for VG2K : 0 – 80 V
6. Multirange Analogue Voltmeter of range : 0 - 5 V, 0 - 15 V , 0 - 100 V
7. Multirange Analogue Ammeter of range : 0 - 1 (100 divisions) with range multiplier : 10^{-7}A, 10^{-8}A & 10^{-9}A

3 Experimental Setup

The experimental setup involves a tube containing low-pressure experimental gas fitted with four electrodes: an electron-emitting cathode (K), a mesh grid (G1) for minimizing space charge effects, a mesh grid (G2) for acceleration, and an anode (A). The anode was held at a slightly negative electrical potential relative to the grid G2 (although positive compared to the cathode) so that electrons had to have at least a corresponding amount of kinetic energy to reach it after passing the grid and thereby making the dips in the plate current more prominent. Instruments were fitted to measure the current passing between the electrodes and to adjust the potential difference (voltage) between the cathode (negative electrode) and the accelerating grids.

4 Theory

The experiment's foundation lies in Bohr's model of quantized energy levels, where electrons can only exist in specific energy states, and transitions between these states occur when an electron absorbs or emits a discrete quantum of energy.

The Franck-Hertz tube in this instrument is a tetrode filled with the vapor of the experimental substance indicating the basic scheme of the experiment. The electrons emitted by the filament can be accelerated by the potential VG2K between the cathode and the grid G2. The Grid G1 helps in minimizing space charge effects. The grids are wire mesh and allow the electrons to pass through. The Plate (A) is maintained at a potential slightly negative with respect to grid G2. This helps in making the dips in the plate current more prominent. In this experiment, the electron current is measured as a function of the voltage VG2K. As Voltage increases, the electron energy goes up and so the electron can overcome the retarding potential VG2A to reach the plate (A). This gives rise to a current in the ammeter, which initially increases. As the voltage further increases, the electron energy reaches the threshold value to excite the atom in its first allowed excited state. In doing so, the electrons lose energy, and therefore the number decreases. This decrease is proportional to the number of inelastic collisions that have occurred. When the VG2K is increased further and reaches a value twice that of the first excitation potential, it is possible for an electron to excite an atom halfway between

the grids, lose all its energy, and then gain a new enough energy to excite another atom resulting in a second dip in the current. The advantage of this type of configuration of the potential is that the current dips are much more pronounced, and it is easy to obtain five-fold or even larger multiplicity in the excitation of the first level i.e. one can get 5 peaks (dips) or more.

5 Methodology

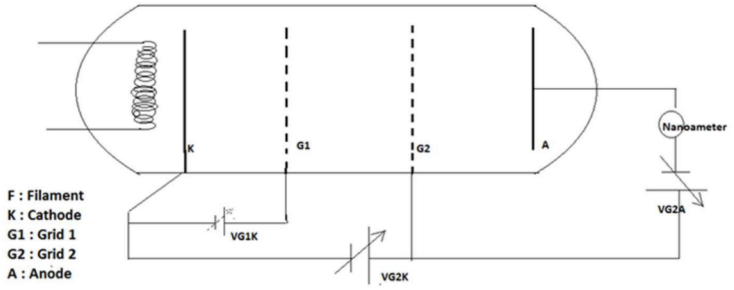


Figure 1: Circuit diagram of Franck-Hertz experiment

- The tube is evacuated to eliminate air molecules that could interfere with electron collisions.
- The cathode is heated to emit electrons, which are then accelerated towards the anode due to the applied electric field.
- The grid electrode allows some electrons to pass through while others experience collisions with Argon atoms. The collector electrode measures the energy distribution of electrons after they pass through the Argon gas.
- As the accelerating voltage increases, plateaus appear in the current-voltage characteristic curve. These plateaus represent instances where electrons are gaining energy corresponding to the quantized energy levels of Argon.
- When the accelerating voltage reaches a value that matches the energy difference between quantized energy levels of Argon, electrons undergo inelastic collisions that result in a noticeable drop in current. This drop signifies that electrons have transitioned to higher energy levels, effectively depleting the current.

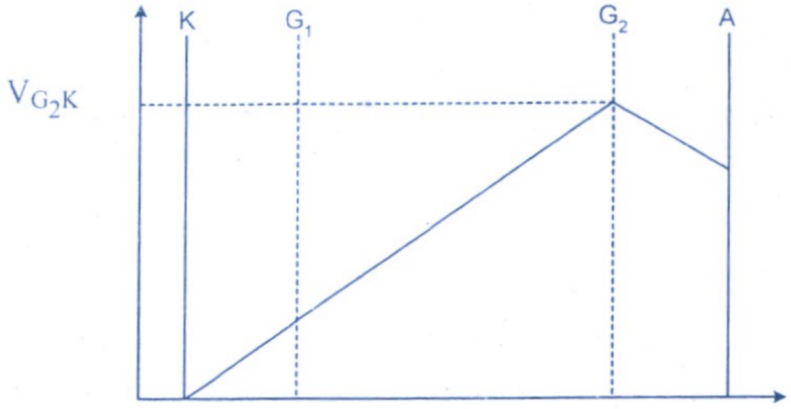


Figure 2: Configuration of the potential in Franck-Hertz experiment

6 Results

VG2K (V)	I (nA)	VG2K (V)	I (nA)	VG2K (V)	I (nA)
8	2	26	16	54	39
9	2	28	23	56	22
10	2	30	25	58	14
11	2	32	18	60	31
12	3	34	8	62	47
13	5	36	13	64	53
14	7	38	28	66	46
15	9	40	36	68	29
16	11	42	32	70	20
17	12	44	18	72	32
18	13	46	8	74	49
20	12	48	21	76	58
22	9	50	41	78	55
24	8	52	47	80	42

Table 1: Data taken for I vs. VG2K characteristics

A graph illustrating the correlation between I and VG2K is plotted to extract information related to the first excitation energy.

The data collection procedure involves measuring at intervals of 1-2 V, while at the dips and peaks. Remarkably, each stepwise increase of approximately 10-12 V corresponds to a notable reduction in current, indicative of an energy transfer of approximately 10-12 eV.

Through analysis of the interpolated curve, an average spacing between peaks of 11.62 V is computed, indicating that the first excitation energy of Argon is around 11.62 eV, a value remarkably close to the theoretical expectation of 11.83 eV.

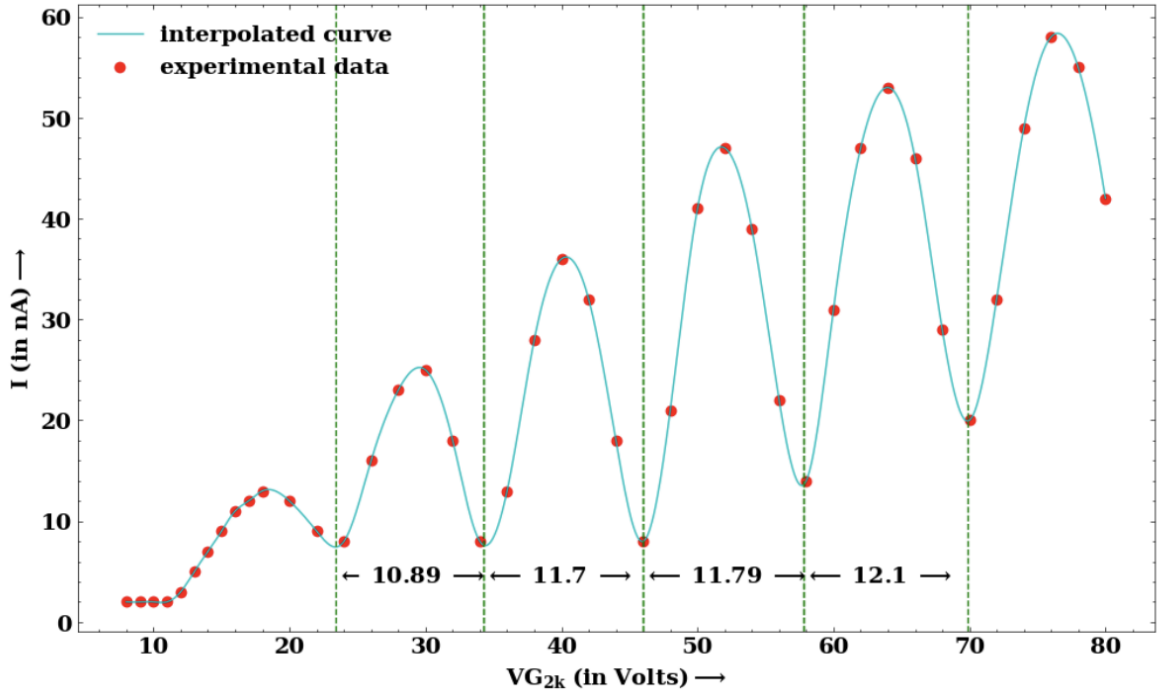


Figure 3: I vs. VG2K characteristics

7 Discussion

The inquiry that naturally emerges during the course of this experimental undertaking pertains to the rationale underlying the selection of Argon as the gas employed within the Franck-Hertz tube, as opposed to alternative gaseous substances. This matter shall be addressed comprehensively within this particular section.

In the context of our experimental objectives, a fundamental necessity resides in the assessment of the excitation energy intrinsic to a specific elemental entity. Consequently, the selection of a gaseous medium becomes contingent upon its monoatomic nature, wherein the stable molecular structure consists of singular atoms. This stipulation inherently excludes a multitude of alternative candidates, including hydrogen and various halogens, on the grounds of their molecular complexity.

Additionally, a crucial criterion pertains to the gaseous state's inherent presence or its facile attainability from a given elemental source. In this context, liquids such as mercury and gallium, along with inert gases, emerge as viable candidates. In fact, the inaugural Franck-Hertz experiment conducted in 1914 employed mercury due to its liquid state at the experimental conditions. Furthermore, the elemental selection necessitates a relatively sizable atomic radius to enhance the likelihood of electron collisions. Argon, characterized as an inert gas possessing a notably substantial atomic diameter relative to inert gases like Helium and Neon, stands as the chosen candidate for our experimental pursuit.

Moreover, the selection process is influenced by empirical considerations encompassing discernible features such as distinct peaks and troughs in the experimental outcomes. Furthermore, the determination is bound by pragmatic factors, including limitations on the voltage that can be applied, ensuring that the excitation energy remains amenable to the constraints of the applied voltage differential. These multifaceted rationales collec-

tively contribute to the preference for one elemental species over its counterparts in our experimental endeavor.

8 Error Analysis

The accuracy of the Voltmeter is 0.1 V and hence I report $\Delta V \sim 0.1$ V as an experimental error in excitation energy determination.

9 Conclusion

Therefore, we have confirmed the quantization of electron energy. Additionally, we present the initial excitation energy of Argon as 11.62 ± 0.1 eV.